Recipe Organizer and Nutrition Calculator

1. Project Overview

Recipe Organizer and Nutrition Calculator is a console-based Python application that allows users to organize recipes, search recipes using ingredients, and analyze nutritional information. The application provides a menu-driven interface where users can add recipes, view stored recipes, search by ingredients, and generate nutrition statistics. This project demonstrates the practical implementation of Python programming concepts through a real-world application.

2. Problem Statement

Develop a Python application that allows users to store and manage recipes efficiently. The application should enable users to:

- Add new recipes
- Store recipe details
- View all recipes
- Search recipes by ingredient
- Analyze nutritional information
- Display recipe statistics
- Handle invalid user inputs gracefully

3. Features

- Add new recipes
- Store recipe name, ingredients, meal type, food type, and calories
- Display all saved recipes
- Search recipes by ingredient
- Calculate total recipes
- Calculate average calories
- Count Vegetarian and Non-Vegetarian recipes

- Display high-calorie recipes (more than 500 calories)
- Display unique ingredients using sets
- Handle invalid user input using exception handling
- Menu-driven application

4. Technologies Used

- Python 3
- VS Code
- Command Prompt / Terminal

5. Python Concepts Used

- Functions
- Lists
- Dictionaries
- Tuples
- Sets
- Loops
- Conditional Statements
- Exception Handling (try-except)
- String Methods
- User Input Validation

6. Program Workflow

Start

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Display Main Menu

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Choose Operation

(Add Recipe | View Recipes | Search Recipe | Analyze Nutrition | Exit)

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Execute Selected Function

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Display Output

↓

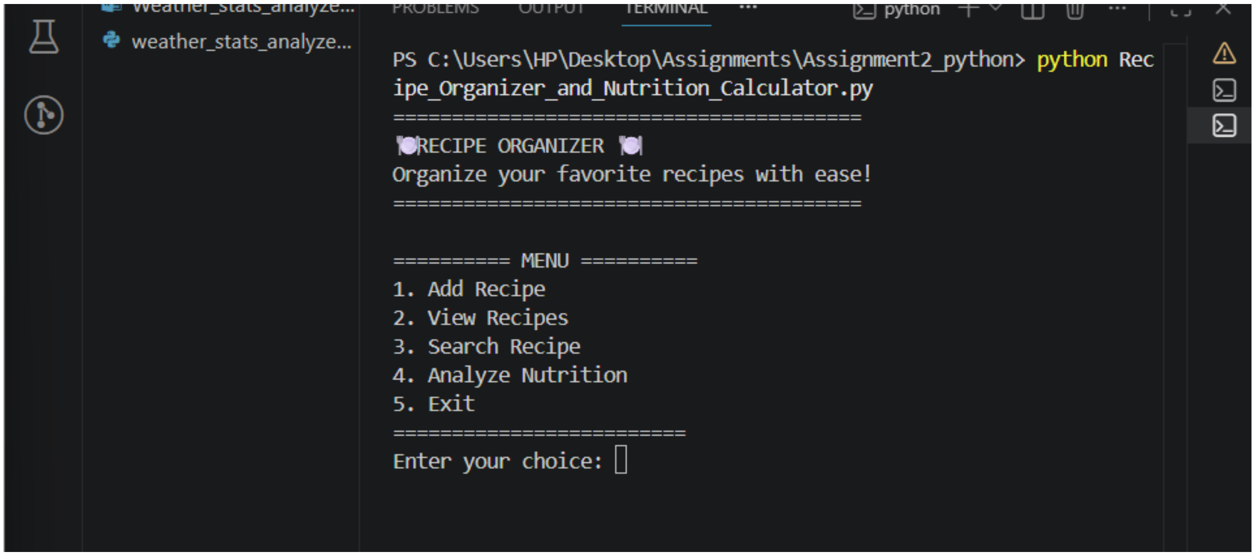
Repeat Until Exit

↓

End

7.Output Screenshots and Explanation

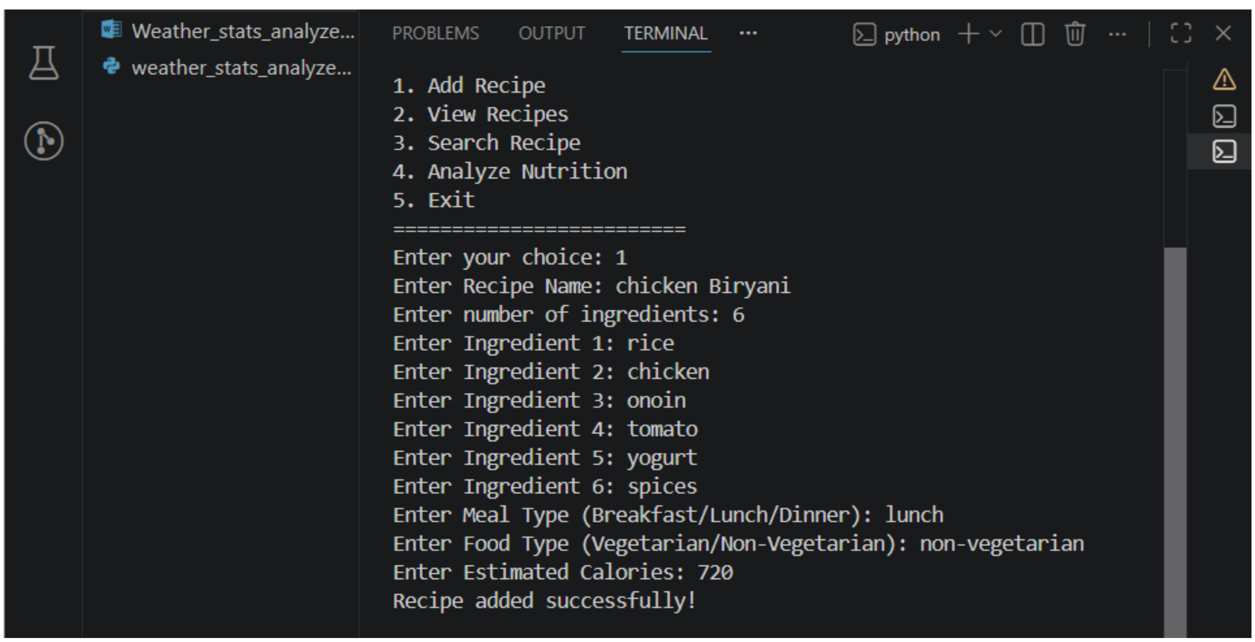
Screenshot 1: Main Menu



Explanation

This screen displays the main menu of the Recipe Organizer application. Users can choose different options such as adding recipes, viewing stored recipes, searching recipes by ingredients, analyzing nutrition, or exiting the application.

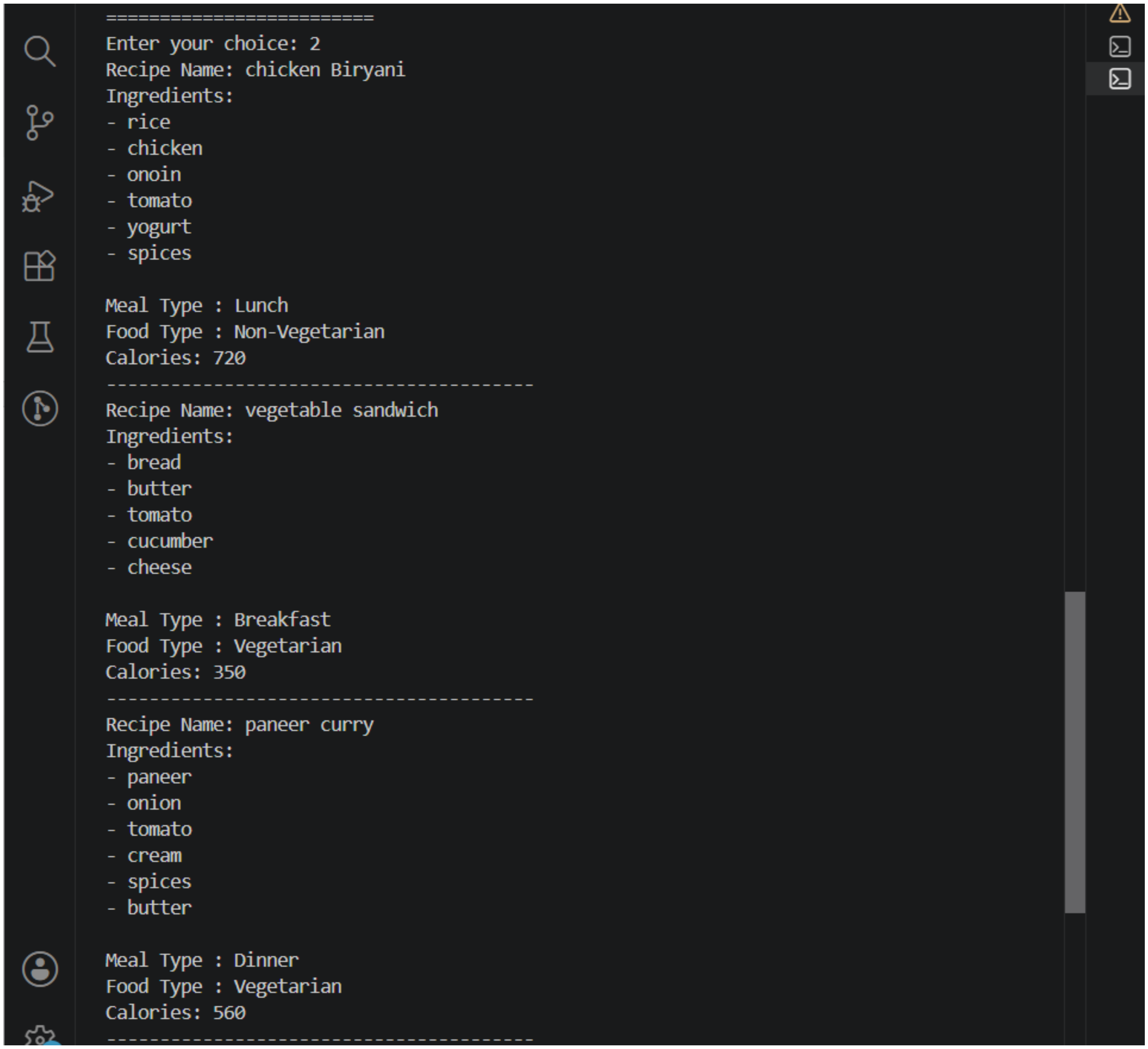
Screenshot 2: Adding a New Recipe



Explanation

This screen demonstrates the process of adding a new recipe. The user enters the recipe name, ingredients, meal type, food type, and estimated calories. The application validates the entered information before storing the recipe successfully.

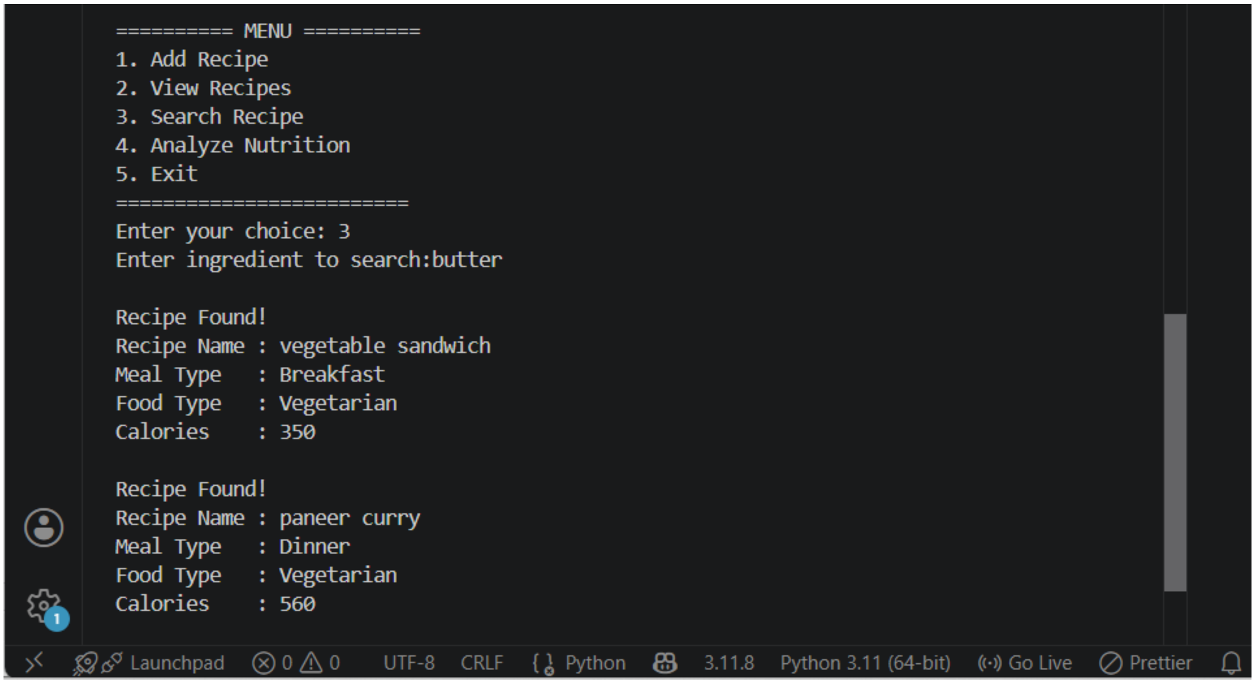
Screenshot 3: Viewing Recipes



Explanation

This screen displays all the recipes stored in the application. Each recipe shows its name, ingredients, meal type, food type, and estimated calories in a well-organized format for easy reading

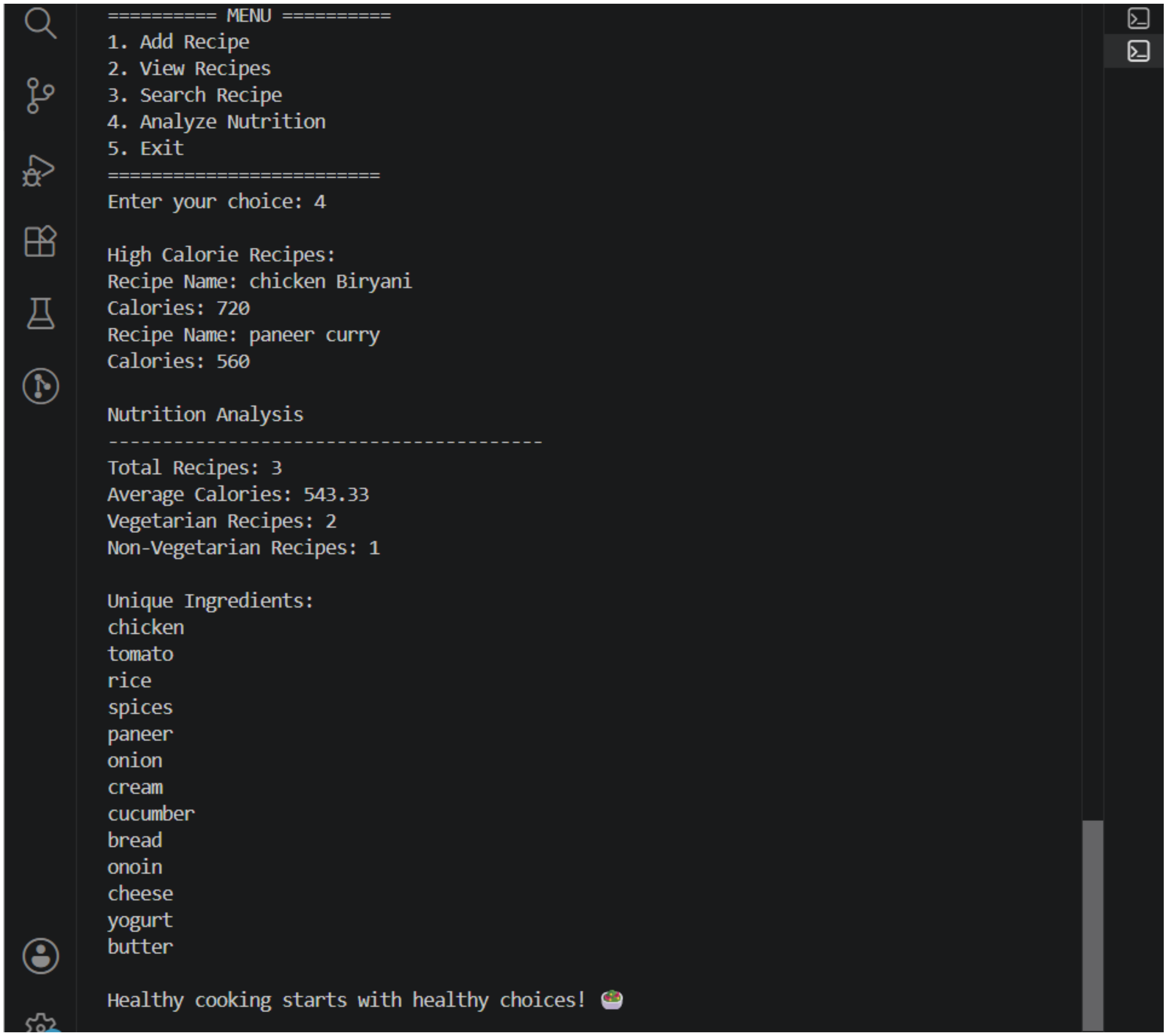
Screenshot 4: Searching Recipe by Ingredient



Explanation

This screen shows the ingredient search feature. The user enters an ingredient name, and the application searches through all stored recipes. If a matching ingredient is found, the corresponding recipe details are displayed; otherwise, an appropriate message is shown.

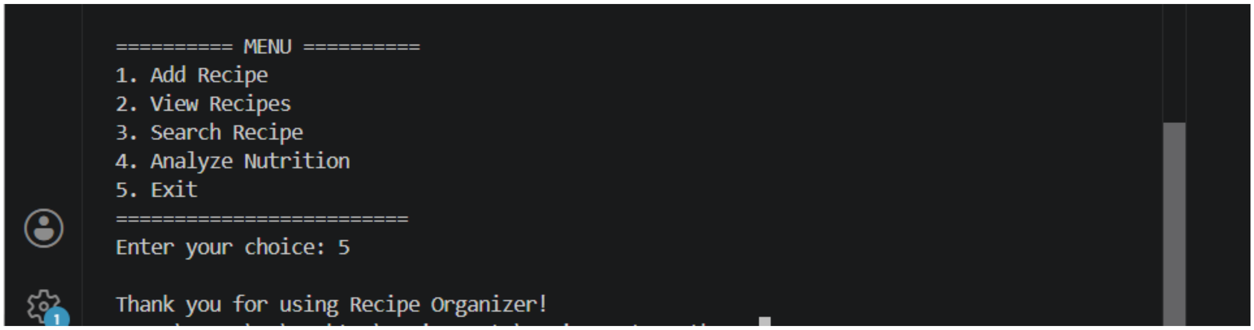
Screenshot 5: Nutrition Analysis



Explanation:

This screen presents the nutritional analysis of the stored recipes. The application calculates the total number of recipes, average calories, counts of Vegetarian and Non-Vegetarian recipes, identifies high-calorie recipes, and displays the unique ingredients used across all recipes.

Screenshot 6: Program Exit



Explanation:

This screen shows the successful termination of the application after the user selects the Exit option. It confirms that the Recipe Organizer has completed execution successfully and thanks the user for using the application.

8. Learning Outcomes

After completing this project, I learned how to:

- Develop a menu-driven Python application
- Organize data using lists, dictionaries, tuples, and sets
- Implement modular programming using functions
- Perform searching and data analysis operations
- Validate user input using exception handling
- Apply Python concepts to solve a real-world problem
- Improve problem-solving and logical thinking skills

9. Conclusion

The **Recipe Organizer and Nutrition Calculator** was successfully developed using Python programming concepts such as functions, lists, dictionaries, tuples, sets, loops, conditional statements, string methods, and exception handling. The application efficiently manages recipe information, performs nutritional analysis, and provides interactive user experience while fulfilling all the project requirements.